

You give them something to eat:

Dietary Diversity, Coverage, and Catholic Aid in Rwanda

Diego González-González*

September 30, 2026

Abstract

How should a food-security organization allocate aid when a trial ranks packages imprecisely and some diets are unobserved? I reanalyze the village-randomized evaluation of Catholic Relief Services' Gikuriro program in Rwanda. At its expected provider budget, lower and large cash have the highest fitted mean when combined in a village lottery. Gikuriro's fitted deficit is 0.237 food groups. I construct coherent dietary distributions, retain information in incomplete food modules, and choose diversified allocations over common regions for all arm means. Diversification lowers worst-case measured loss relative to restricting the choice to budget vertices. Its magnitude depends strongly on inference. A conditional finite-assignment outer region gives a mean-loss bound of 9.440 groups; an exploratory block approximation gives 0.570. Synthetic tests under the released quota structure show substantial undercoverage of that approximation. Neither calculation supplies a reliable small-loss recommendation. The result separates the arithmetic appeal of concentrated cash from the evidence required to defend an allocation for the weighted baseline sample. Deployment elsewhere additionally requires stable costs, delivery, and spillovers. The contribution is an empirical decision analysis of an existing experiment, rather than new identification or a statistical minimax-regret rule.

Keywords: food security; Catholic Relief Services; cash transfers; dietary diversity; treatment allocation.

JEL codes: C93, D61, I32, O12.

*Research draft, September 2026. This paper uses public data from an experiment conducted by Craig McIntosh and Andrew Zeitlin, with Innovations for Poverty Action and the implementing organizations. It does not report a new field experiment. Code and drafting assistance were provided by OpenAI Codex; the draft requires the author's review before journal submission. No institutional affiliation, funding, or endorsement is asserted. Replication materials: <https://github.com/dgonzalezgonzalez/you-give-them-something-to-eat>.

1 Introduction

A food-security organization must decide both what to provide and how many households to reach. Giving a large transfer to a small share of villages can improve recipients' diets while leaving most villages untreated. A less expensive package can reach everyone and produce a smaller gain per recipient. An experimental comparison of package means informs this choice only after the untreated households, the provider budget, and uncertainty about the outcomes enter the same calculation.

I examine this question in the Gikuriro program in Rwanda. Catholic Relief Services (CRS) and SNV implemented the program with support from USAID. Nutrition and health activities, agricultural support, saving groups, and water, sanitation, and hygiene interventions formed a single package. Its stated aim was to improve maternal and child nutrition. The experiment of [McIntosh and Zeitlin \(2024\)](#) assigned 248 villages to Gikuriro, control, or one of four cash amounts delivered by GiveDirectly. The title recalls Mark 6:37. Religious affiliation was not randomized, so the experiment cannot identify a specifically Catholic delivery mechanism.

The parent study already develops equal-cost cash benchmarking, cost effectiveness, and the tradeoff between coverage and intensity. For mean diet, the Gikuriro comparison against a control/large-cash lottery is exactly its cost-effectiveness contrast multiplied by Gikuriro's cost. Its studentized statistic is unchanged. I therefore ask a narrower question: what allocation can be defended when the objective includes the diet distribution, missing outcomes are allowed to differ by assignment, and the organization can diversify across packages? The economic distinction is between the fitted winner and the opportunity cost that the evidence can credibly bound.

At the source's expected provider budget of \$124.49 per eligible household, large cash alone is unaffordable. A control/large lottery serves 24.1 percent of villages. A lower/large lottery serves every village with cash and reserves the large package for 15.3 percent. The latter has the highest fitted mean dietary diversity. Choosing Gikuriro instead has fitted loss 0.237 groups on the twelve-group household score. Missing-outcome endpoint means, however, allow the sign of this contrast to reverse. The stronger recipient-level response to large cash does not resolve the equal-budget decision.

Uncertainty also changes the relevant policy class. A linear objective with known arm means has an optimum at a budget vertex. Minimizing worst-case regret over a region for those means need not have that property. I allow the chosen allocation to use every observed package, while the comparator is the best feasible allocation for each possible vector of arm means. For the weighted baseline mean, diversification reduces the exploratory block-region bound from 0.908 to 0.570 groups. It includes lower, upper, and large cash, a small Gikuriro share, and some control. This is a calculation under an estimated region, rather than a recommended portfolio.

The distinction matters because the inference approximation is unreliable in finite-block stress tests. Fixed treatment counts induce dependence across arms within each of the 22 assignment blocks. I retain that covariance, use a common block multiplier, guard zero estimated variances by deterministic support, and account for simulation error in the multiplier quantile. Even after these

repairs, joint coverage in synthetic fifteen-comparison tests is 0.869 with equal weights and 0.740 with unequal frame weights. Those experiments falsify a general nominal-coverage interpretation of the approximation used here. They do not measure the unknown actual coverage of the Rwanda estimates.

I consequently report a separate finite conditional-assignment outer region. It uses bounded outcomes, fixed baseline weights, independent blocks, and exchangeable village labels conditional on each released block’s counts. The resulting diversified mean-loss bound is 9.440 groups. It is too wide to support a small-loss decision. Its width does not prove that every valid inferential procedure must be uninformative or that additional data are indispensable. It establishes the practical limit of this deliberately conservative calculation under stated assumptions. Representing unsampled households or changing program saturation requires further assumptions beyond the conditional assignment event.

The decision problem draws directly on [Manski \(2004, 2007\)](#) and [Stoye \(2007\)](#). Their work makes missing outcomes and multiple treatments part of treatment choice. [Kitagawa and Tetenov \(2018\)](#) studies empirical welfare maximization in richer policy classes. Here I minimize a confidence-region bound after observing the data. I do not establish a rule that minimizes worst-case expected regret over repeated samples. The contribution is the empirical accounting of diversification, selection, and inferential scope in an important cash-benchmarking experiment. The underlying treatment-choice logic is established; the empirical question is how much protection it supplies at the trial’s budget and inferential support.

The measured objective remains limited. Household dietary diversity describes current food access, while child nutrition, future resources, and public services may rank aid differently. [Currie and Gahvari \(2008\)](#) reviews reasons for in-kind provision; [Banerjee and Duflo \(2007\)](#) documents spending priorities beyond food; and [Haushofer and Shapiro \(2016\)](#) studies cash effects on consumption, assets, and food security. The local price and nonrecipient responses in [Cunha et al. \(2019\)](#); [Egger et al. \(2022\)](#) further limit transport of an original-delivery lottery. An illustrative saving model in the appendix explains how intertemporal utility can rise without a current dietary gain. The bundled experiment supplies no variation identifying that mechanism separately.

2 Decision problem and theoretical framework

Let $a \in \mathcal{A} = \{0, g, \ell, m, u, L\}$ denote control, Gikuriro, and the four cash packages. Let $D_i(a)$ be the diet of baseline household i when its village receives assignment a , at the original within-village saturation. A higher value of a specified transformation $v(D)$ is preferred. I consider the mean $v(d) = d$, all integer survival indicators, and negative linear shortfalls

$$v_z(d) = -\frac{(z - d)_+}{z}, \quad z = 1, \dots, 12. \quad (1)$$

The thresholds are alternative priorities over a household food-access proxy. They are not official hunger cutoffs. The shortfall structure follows [Foster et al. \(1984\)](#); it does not assign nutritional

equivalence to all food groups.

For fixed positive baseline weights w_i , define $\mu_a(v) = \sum_i w_i v(D_i(a)) / \sum_i w_i$ over the released baseline eligible sample. This target conditions on sample inclusion and its weights. Treating it as the mean of the entire eligible frame additionally requires the household sampling and observation assumptions. For policy deployment, I consider comparable villages with the same eligible-household weighting and package delivery. That transport restriction is maintained, not tested by random assignment.

A village lottery chooses $q_a \geq 0$ with $\sum_a q_a = 1$, independently of potential outcomes conditional on target blocks. With fixed provider costs c_a and $c_0 = 0$, the feasible set is

$$\mathcal{Q}(b) = \{q \geq 0 : \mathbf{1}'q = 1, c'q \leq b\}. \quad (2)$$

The source costs are standardized average costs per eligible household. In particular, cash overhead is costed separately as if each package served 56,127 households across eight districts, a national scale comparable to Gikuriro. These accounts do not observe the marginal cost of a small package share. Equation (2) conditions on their proportional allocation and stable delivery. It imposes an expected cost per eligible household; it does not ensure an exact realized cap for finitely many villages of unequal size.

Proposition 1 (Observed-package lotteries). *Under stable package outcomes and no cross-village interference, a lottery has $F_q(d) = \sum_a q_a F_a(d)$ and value $q'\mu(v)$. For known $\mu(v)$, a maximizer over $\mathcal{Q}(b)$ exists with at most two packages in its support.*

When $c_a < b < c_h$, the binding two-package vertex has

$$q_h = \frac{b - c_a}{c_h - c_a}, \quad q_a = 1 - q_h. \quad (3)$$

Every pure feasible package is also considered, including policies that leave funds unspent. At $b = c_g$, the full six-package set has nine vertices: pure Gikuriro and eight cash/control vertices. The latter are control, the three small pure cash packages, and four cheaper-package/large-cash mixtures. A budget change requires recomputing the vertices, including cost-order reversals.

Corollary 1 (Coverage comparison). *Universal Gikuriro exceeds the control/large lottery for objective v precisely when*

$$\mu_g - \mu_0 \geq (c_g/c_L)(\mu_L - \mu_0).$$

For mean diet, the difference equals $c_g\{\tau_g/c_g - \tau_L/c_L\}$, the rescaled parent-study cost-effectiveness comparison.

For uncertain arm means in a nonempty compact region $\mathcal{C}$, define

$$R(q, \mu) = \max_{r \in \mathcal{Q}(b)} (r - q)'\mu, \quad U_{\mathcal{C}}(q) = \sup_{\mu \in \mathcal{C}} R(q, \mu). \quad (4)$$

Regret measures lost expected value in the specified diet objective. If $\mathcal{C}$ covers the common true means, $U_{\mathcal{C}}(q)$ bounds their regret. A selected allocation may therefore use the same event without selecting a separate test after fitting it.

Proposition 2 (Diversification over a common region). *Let $\mathcal{V}$ be the vertices of $\mathcal{Q}(b)$. Then*

$$U_{\mathcal{C}}(q) = \max_{r \in \mathcal{V}} \sup_{\mu \in \mathcal{C}} (r - q)' \mu$$

is convex in q . A minimizer over $\mathcal{Q}(b)$ exists and need not be a vertex. If $\mathcal{C} = \{\mu : A\mu \leq d\}$ is compact, the minimum is obtained from a linear program with a nonnegative dual vector for each comparator vertex. On a simultaneous coverage event, its selected allocation has regret at most its reported bound.

For example, two treatments with costs below budget and independent mean intervals $[0, 1]$ have worst-case loss $\max(q, 1 - q)$ when q is the share given the first. The best vertex has loss one and the equal mixture has loss one half. Proposition 1 therefore limits the comparator under known means; it does not restrict the choice minimizing (4). This elementary illustration is established decision logic, not a new theorem about treatment allocation.

The fitted maximizer uses $\hat{\mu}$ as if known. The region minimizer instead protects against every mean vector admitted by $\mathcal{C}$. Neither procedure here is proved statistical minimax regret in the repeated-sampling sense. The region is an outer set rather than the sharp set of all compatible potential outcomes and distributions. Its bound can be conservative even when its coverage is valid.

3 Setting, sources, and measurement

The baseline survey took place in August–September 2016 and endline in September–October 2017. Gikuriro included village nutrition schools, farmer field learning schools, agricultural inputs, small livestock, Savings and Internal Lending Communities, and environmental health promotion. Eligibility combined poverty classification with a young child or a pregnant or lactating woman. Cash assignment operated at village level and covered the source’s eligible beneficiaries. Assignment effects include receipt, nonreceipt, and within-village consequences under original delivery.

The parent paper describes computer randomization in 22 district/poverty blocks of 10–13 villages. The released counts give 74 Gikuriro, 74 control, 22 each lower, middle, and upper cash, and 34 large-cash villages. Each small cash arm has one village in every block. The inspected public replication scripts contain no original assignment program establishing every randomization restriction. I assume labels are exchangeable conditional on the released block counts and assignments are independent across those blocks. Uniform quota assignment implies this model; additional assignment restrictions need not. The probabilities used below are $p_{ba} = n_{ba}/n_b$ under that explicit conditional model, rather than known solely because realized frequencies can be calculated. The public registry and its count discrepancy are documented in the replication design note; they do not resolve this missing assignment information.

Table 1: Village assignment, household observation, and provider costs

Arm	Cost (USD)	Villages	Baseline eligible	Observed diet	Retained (%)
Control	0	74	521	497	96.0
Gikuriro	124.49	74	540	522	97.4
Lower	53.58	22	165	160	96.4
Middle	95.86	22	154	148	97.3
Upper	121.24	22	167	164	98.9
Large	517.44	34	246	237	96.9

Notes: Costs are source USD per eligible household under the standardized accounting convention. Baseline weights expand the released sample; complete-module endline observation is distinct from item-informed diet identification. No treatment-receipt conditioning is used.

I use the corrected CC BY 4.0 release of [McIntosh \(2025\)](#), DOI [10.5281/zenodo.15881329](#). The earlier deposit was restricted after identifying fields were found. Its replacement explicitly reports their removal. The archive matches the provider’s checksum. The public numeric extract uses arbitrary replacement household, village, block, and child keys; it excludes contacts, geographic labels, dates, and beneficiary lists. These are processed research data. I do not reconstruct all upstream survey cleaning.

The household extract contains 5,506 rows. There are 1793 baseline eligible households, compared with 1,794 in the original paper, and 1751 eligible endline panel households. Complete endline food modules identify 1728 integer diets. Item-level information identifies two further diets, giving 1730. Table 1 reports the released assignment and initial complete-module sample; Table 12 accounts for the full baseline cohort.

HDDS counts twelve groups consumed by anyone in the household during the preceding day. The questionnaire’s sixteen categories are combined into twelve by taking any positive vegetable, fruit, or meat component for the respective group. Other groups remain distinct, including oils, sweets, and condiments. This follows the household score described by [FAO \(2011\)](#). It measures food access and variety, with a ceiling of twelve; it cannot establish calorie sufficiency, individual nutrient adequacy, or hunger severity.

The supplied endline score includes seven noninteger values near 4.463. Underlying responses can be missing or contain 888/999 codes. Complete binary modules match the reconstructed integer score exactly. For an incomplete module, a combined group is certainly positive if any component is positive and certainly zero only if every component is observed zero. The remaining uncertain groups define an interval $[D_i^-, D_i^+]$. Absent endline diets receive $[0, 12]$. The reconstruction retains this information instead of placing an imputed fractional score into an integer distribution. Source scores remain in the crosswalk and secondary sensitivity.

The supplemental household file provides source consumption, productive assets, saving, borrowing, health knowledge, sanitation practices, and purchased/own-produced food values. Economic source indices and inverse hyperbolic sine (IHS) transformations are inherited where provided. Both food-value fields use source weekly recall and a newly computed asinh transformation;

the inspected labels do not recover a defensible monetary price base for conversion. I report transformation units rather than invented dollars or percentage changes. The individual panel provides standardized height-for-age, weight-for-age, and upper-arm scores, ages, and anthropometry flags. Those flags permit the child cohort accounting below. Every auxiliary outcome remains exploratory.

4 Empirical strategy and inferential scope

4.1 One distribution for each arm

For an identified transformation $Y_i = v(D_i)$ and its observation indicator R_i , define

$$\hat{\mu}_a = \frac{\sum_i T_{j(i),a} w_i R_i Y_i / p_{b(i),a}}{\sum_i T_{j(i),a} w_i R_i / p_{b(i),a}}. \quad (5)$$

The weights are positive and fixed at baseline. They numerically equal the eligible village frame count divided by its released baseline sample count, within floating-point precision. I invent no intensive-tracking factor. For identified diets, (5) targets the weighted potential-observation ratio under each arm. Different assignment-specific observation can change its composition. An observed-diet policy calculation is consequently a sensitivity measure; it represents policy diet for the baseline target only with an additional selection assumption.

For the baseline outcome, I replace Y_i by $v(D_i^-)$ or $v(D_i^+)$ and set $R_i = 1$ for every baseline eligible. Since every reported v is increasing, these endpoint means bound the common true arm mean. A contrast uses lower endpoints for positive coefficients and upper endpoints for negative coefficients to form its lower bound, with the assignments reversed for its upper bound. The resulting estimated endpoint region is subject to sampling uncertainty. It is not already a confidence interval for missing diets.

Applying (5) to indicators gives a normalized empirical CDF. Its means and shortfalls satisfy

$$\widehat{\mathbb{E}}[D] = \sum_{k=1}^{12} \widehat{\Pr}(D \geq k), \quad \widehat{\mathbb{E}}[(z - D)_+] = \sum_{k=0}^{z-1} \widehat{\Pr}(D \leq k)$$

exactly. The mean, twelve survivals, and twelve linear shortfalls have two affine duplicates, leaving 23 distinct transforms. The analysis covers increasing and increasing-concave priorities on the bounded integer score. Ratio normalization ensures coherence; it does not make finite-sample ratios exactly unbiased.

4.2 Finite conditional-assignment outer regions

Let $W_j = \sum_{i \in j} w_i$ be the fixed eligible baseline weight in village j , $W_b^{\max} = \max_{j \in b} W_j$, and $k_{ba} = n_{ba}$. For a primitive transformation supported on $[L, H]$, let $r = H - L$ and

$$V_a = \sum_b k_{ba} \left(\frac{W_b^{\max}}{p_{ba}} \right)^2, \quad t_a = r \sqrt{\frac{1}{2} V_a \log(2M/\alpha)}. \quad (6)$$

Here M is the number of primitive arm/endpoints protected together. If $\hat{D}_a$ is the denominator of (5), the interval $[\hat{\mu}_a - t_a/\hat{D}_a, \hat{\mu}_a + t_a/\hat{D}_a] \cap [L, H]$ covers its fixed conditional target with probability at least $1 - \alpha/M$. A zero denominator gives full support. Appendix A proves this through bounded sampling without replacement using Hoeffding (1963).

I use $M = 276$ for six arms, 23 transforms, and two full-baseline endpoints, with $\alpha = 0.05$. A separate identified-outcome family uses $M = 138$. These are separate 95 percent events. Their conjunction is not asserted to have 95 percent coverage. The primitive arm box supplies a common region for all policy comparisons and diversified allocations within a given family. For an unobserved diet, the lower endpoint’s lower confidence bound and upper endpoint’s upper bound enclose the true mean. This finite statement conditions on the exchangeable assignment model, the fixed weights, and stable potential outcomes. It does not prove full-frame sampling representativeness, transport, or a hard budget guarantee.

4.3 Exploratory block approximation and stress tests

For block b and arm a , let N_{ba} and D_{ba} be the numerator and denominator contributions in (5). The ratio score is

$$u_{ba} = \frac{N_{ba} - \hat{\mu}_a D_{ba}}{\sum_h D_{ha}}, \quad \hat{\Sigma} = \frac{B}{B-1} \sum_b u_b u_b', \quad B = 22. \quad (7)$$

Grouping before covariance retains dependence among all arms assigned within a block. Arm-exclusive village scores would force cross-arm covariance to zero. Appendix A gives an exact counterexample: that omission estimates only 90 percent of the true difference variance in a ten-village quota design, even as the number of identical blocks increases. The common block covariance repairs this omission. Its asymptotic justification additionally requires increasing independent blocks, stable positive denominator limits, and no dominating block.

Shared block Rademacher multipliers generate exploratory maximum-studentized bands with 99,999 draws. A 99 percent Monte Carlo upper order statistic protects the estimated 95 percent multiplier quantile from simulation error within each family. That Monte Carlo protection does not address sampling undercoverage. I retain separate 184 Gikuriro/cash, 828 all-policy, and 1656 endpoint families, deduplicating affine repetitions. A zero estimated variance receives its deterministic support range rather than a structural-zero interval; 121 displayed rows require this guard. Pointwise intervals use 21 block degrees of freedom. None of these choices establishes exact weak-null inference.

Synthetic tests use the actual released quota structure, 800 repeated assignments per scenario, and a 4,999-draw diagnostic multiplier calculation for fifteen mean comparisons. Table 2 reports joint coverage. The equal-weight and unequal-weight failures remain material even with Monte Carlo upper quantiles. The rare-effect scenario obtains full-support intervals when its realized variance vanishes. Its full coverage cannot rescue the other scenarios. I therefore label every resulting block-region allocation and loss bound as exploratory. ANCOVA CR2 diagnostics concern

Table 2: Finite-block stress diagnostics for the exploratory approximation

Scenario	Covered	Replications	Joint coverage	MC lo.	MC hi.
balanced equal weights	695	800	0.869	0.843	0.891
block heterogeneity	695	800	0.869	0.843	0.891
unequal frame weights	592	800	0.740	0.708	0.770
rare positive effect	800	800	1.000	0.995	1.000

Notes: Synthetic fixed potential outcomes, independent exchangeable assignments, and the 22 released block quotas. Coverage protects fifteen arm-pair mean contrasts, rather than the actual 25-transform endpoint family. MC intervals are exact binomial intervals for simulated coverage. They are neither actual Rwanda coverage estimates nor evidence that nominal 95 percent coverage has been attained.

Table 3: Mean dietary comparisons for the weighted baseline sample

Comparison	ID lo.	ID hi.	Approx. lo.	Approx. hi.	Finite lo.	Finite hi.
Large vs. control	0.171	0.731	–	–	-11.257	12.000
GK vs. Control + Large	-0.275	0.323	-0.703	0.686	-11.436	11.567
GK vs. Lower + Large	-0.574	0.101	-1.113	0.684	-12.000	11.567
GK vs. Upper + Large	-0.293	0.005	-0.771	0.437	-12.000	11.567

Notes: Units are food groups. Positive values favor the first policy. ID columns are estimated missing-diet endpoint regions, not confidence limits. Approximate columns are the exploratory block endpoint bands, which failed stress diagnostics. Finite columns derive from the simultaneous 276-primitive conditional-assignment family. Large cash is unaffordable as a pure equal-budget policy, so its direct comparison is not included in the nine-policy multiplier family; no such approximate band is invented.

a different estimator and cannot validate these ratios or missingness endpoints.

5 Results: diet comparisons and feasible allocations

5.1 Assignment contrasts allowing incomplete diets

Large cash has a higher observed mean than control. The initial ANCOVA estimate is 0.545 groups, and the reconstructed parent specification reproduces the original paper’s published rounding (Appendix Table 13). That is an assignment comparison among measured outcomes. In the full weighted baseline sample, the item-informed estimated region for large cash minus control is $[0.171, 0.731]$ groups. Its positive sample endpoints are consistent with a full-baseline improvement, but the finite conditional envelope $[-11.257, 12.000]$ is wide and includes zero. I consequently do not claim an established positive full-baseline effect from these bounds.

For the feasible budget comparisons in Table 3, Gikuriro’s estimated mean region crosses zero against each reported cash lottery. Against lower/large cash, the observed-diet ratio difference is -0.237 groups. The sign remains unresolved after allowing incomplete diets and inferential uncertainty. A lottery over observed packages makes the untreated fraction explicit; it does not identify the outcome of giving every household a transfer equal to the lottery’s average cash amount.

One common diet distribution also permits comparisons across every survival threshold and

Table 4: Worst-case measured loss over common arm-mean regions

Objective	Region	GK	Fitted cash	Best vertex	Diversified
Mean HDDS	Finite conditional	12.000	12.000	12.000	9.440
Mean HDDS	Block sensitivity	1.172	0.908	0.908	0.570
Shortfall ($z = 6$)	Finite conditional	0.774	1.000	0.770	0.715
Shortfall ($z = 6$)	Block sensitivity	0.124	0.113	0.099	0.055

Notes: All rows allow arbitrary missing diets consistent with observed items. Mean loss is in food groups; shortfall loss is in normalized objective units. Finite rows use the proved conditional-assignment outer box. Block rows use an explicitly unvalidated approximation and supply no established 95 percent loss guarantee. “Fitted cash” is lower/large cash selected using identified-diet means. “Best vertex” minimizes the same bound over nine vertices; “Diversified” minimizes it over the entire feasible set.

linear shortfall. The estimated six-group shortfall endpoint comparisons can favor cash, while their uncertainty envelopes include zero. Neither increasing nor increasing-concave dominance is established by the finite region. Oils, sweets, and condiments remain components of the score, so even a uniform distributional ranking would concern measured household food access rather than individual nutrition.

5.2 Fitted choice and diversification

Lower/large cash has the highest fitted mean across the nine original-budget vertices. It raises the allocation mean relative to Gikuriro by 0.237 groups in the identified-diet comparison. The fitted winner varies little across initial regression adjustments, but that agreement cannot establish its population optimality. Uncertainty is about the arm means that determine the whole feasible frontier.

Table 4 compares the worst-case loss of universal Gikuriro, the fitted cash vertex, the best vertex for each region, and the unrestricted diversified allocation. In the finite conditional region, the best mean vertex has bound 12.000 groups, while diversification lowers it to 9.440. That numerical improvement is substantial relative to the twelve-group support but still far too large to support a small-loss recommendation. For the normalized six-group shortfall, the diversified finite bound is 0.715 on a unit-range objective.

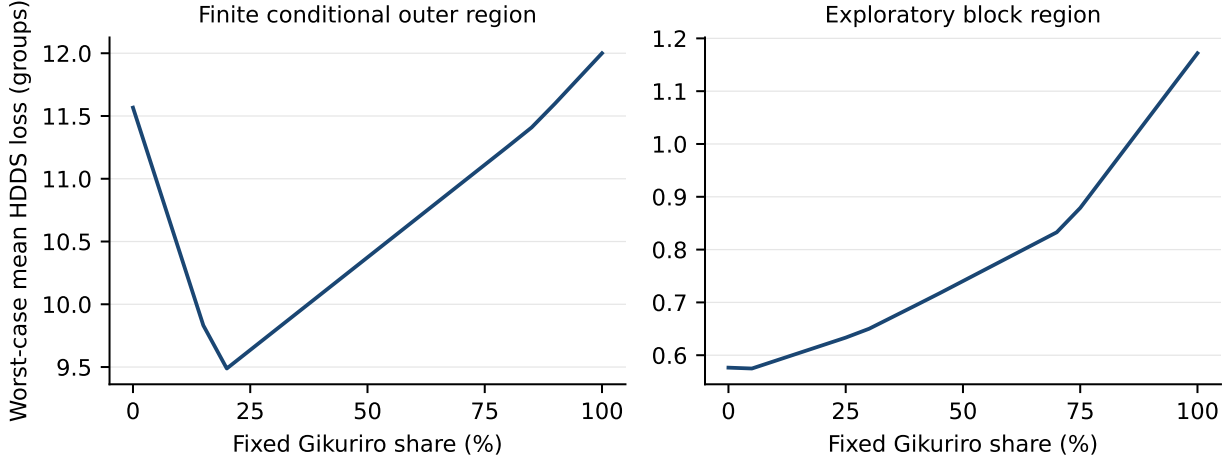
The block sensitivity gives a much smaller diversified mean bound, 0.570 groups, compared with 0.908 for its best vertex. It also illustrates why restricting the chosen policy to vertices can exaggerate attainable precision requirements. The former is conditional arithmetic under an approximation that has failed coverage diagnostics. I withdraw a reliable one-group or half-group certification interpretation. Failure of this particular finite outer region to reach those tolerances likewise cannot prove that the data necessarily preclude all sharper valid decisions.

Table 5 makes the diversification concrete. The exploratory mean allocation places about 44 percent in lower cash, 36 percent in upper cash, 10 percent in large cash, 3 percent in Gikuriro, and 6 percent in control. The finite-region solution instead spreads assistance across all packages and leaves resources unspent. Its diffuse shares largely reflect the broad region. They do not reveal

Table 5: Mean-loss-minimizing village allocations

Region	Control	Gikuriro	Lower	Middle	Upper	Large	USD
Finite conditional	16.1	18.4	21.3	21.3	21.3	1.5	88.32
Block sensitivity	6.1	3.4	43.8	0.0	36.5	10.2	124.49

Notes: Shares are percentages and sum to 100 up to rounding. Expected expenditure is USD per eligible household. The optimizer may leave the budget partly unspent; independent primal adversaries verify each linear-program certificate. These allocations illustrate region-dependent loss protection and are not recommendations for deployment.

**Figure 1:** The measured-loss frontier at a fixed Gikuriro share

Notes: Separate vertical scales make the finite and approximate regions visible. A fixed Gikuriro share can represent a universal-contact restriction only at 100 percent; intermediary shares are village lotteries. The block-region curve has no validated nominal-coverage interpretation. Costs and original-delivery outcomes are fixed.

household preferences or provide evidence that every intervention has a positive effect. Figure 1 traces the same loss calculation while fixing the Gikuriro share. Each point reoptimizes the other package probabilities.

5.3 Cost accounting and magnitude

The simplest back-of-the-envelope calculation is reach: the \$124.49 budget finances only 24.1 percent large-cash coverage at its \$517.44 standardized cost. Combining lower with large cash instead uses 15.3 percent large packages and reaches every village. These shares follow from source costs and the lottery definition. They do not assume a linear household response to dollars. Changes in prices, implementation scale, or participation may alter both costs and outcomes.

I rebuild the policy set in 32 accounting scenarios: a five-by-five grid of Gikuriro and cash cost factors, three common-take-up settings, and four Gikuriro avertable-cost settings. The source decomposition is $c_a^{elig} = c_a^{benef}(1 - s_a + s_a r_a)$, where s_a is the avertable share and r_a compliance. The scenarios use coherent Hájek mean point estimates and the same common arm-mean regions as the main allocations. They no longer substitute initial ANCOVA means for the headline decision. Outcomes remain fixed as costs change, making these accounting sensitivities rather than treatment-

Table 6: Diversified mean-loss bounds under selected cost scenarios

Accounting scenario	Fitted cash	Upper-cost margin (USD)	Finite loss	Approx. loss
GK costs -25%	Lower + Large	-27.88	8.838	0.577
GK costs $+25\%$	Lower + Large	34.37	9.449	0.571
Cash costs -25%	Lower + Large	33.56	9.452	0.572
Cash costs $+25\%$	Lower + Large	-27.06	9.014	0.578

Notes: Every scenario reenumerates vertices and optimizes all six allocation shares. The upper-cost margin is budget minus pure upper-cash cost; a negative value removes that pure package from the feasible set. Loss is in food groups. The full scenario file retains every case. Approximate bounds remain exploratory.

delivery forecasts.

Upper cash costs only \$3.25 less than the original budget, so relative-cost changes can remove its pure vertex. This affects the fitted observed policy and its competitors even before uncertainty enters. Genuine fixed activation costs, $F_a \mathbf{1}\{n_a > 0\} + m_a n_a$, create a different feasible set. The nationally standardized average-unit accounts do not identify those costs for a small mixed deployment. I therefore condition on the stated allocated-cost convention rather than invent a hard-budget solution.

6 Robustness and interpretation

6.1 Regression crosswalk and weight sensitivity

The appendix retains the original-style ANCOVA, its frozen early analysis family, and the coarse missing-outcome crosswalk. The regression conditions on baseline transformations, a missing indicator, and assignment blocks. It imputes missing baseline controls by the weighted within-block mean and never fills a missing endline outcome for the integer analysis. Village CR1 standard errors and maximum-studentized bands are working asymptotic assessments. CR2/Satterthwaite corrections use a specified inverse-weight working covariance. These estimators have different scopes from the block ratio and finite conditional calculation.

The source-pooled crosswalk reproduces the original study’s rounded Gikuriro and large-cash mean coefficients. Subsequent columns split the smaller cash arms, remove the original source-selected control list, and change the outcome and baseline measurement. No new LASSO selection is run. Equal-household and equal-village specifications also change which households or villages receive representation. Their signs provide composition sensitivity rather than independent estimates of the same weighted target. Weight concentration and village/block omission diagnostics expose influential units without making the remaining ratios selection robust.

6.2 Baseline child cohort and auxiliary outcomes

The endline anthropometry sample combines children with different cohort histories. Source flags identify 2265 baseline-eligible children measured at baseline. Of these, 2213 have a linked endline

Table 7: Child cohort accounting under source anthropometry flags

Arm	Baseline cohort	Linked endline	Measured endline	All endline due	Outside cohort
Control	676	654	628	893	239
Gikuriro	672	659	650	913	254
Lower	199	193	189	262	69
Middle	196	190	187	253	63
Upper	204	204	199	287	84
Large	318	313	303	409	97

Notes: Baseline cohort membership is pre-treatment: eligible households and source baseline-anthropometry flag. Linked and physically measured endline counts differ from nonmissing standardized follow-up scores. “Outside cohort” concerns membership in this measured cohort, not identified births. Assignment and weights are inherited from baseline households.

row. The source endline eligibility/due flags identify 3017 children, of whom 2211 belong to that baseline cohort and 806 do not. The latter are outside the baseline-measured cohort; the flags do not show that every one is a newborn. Table 7 reports the partition by assignment.

To assess cohort composition, I retain those baseline children regardless of endline eligibility or due status and estimate follow-up score regressions using their baseline assignment, weights, and source lagged scores. Missing follow-up scores still select observations. No estimate in the separate fifteen-test Holm family is significant at five percent (Appendix Table 11). This does not identify a zero full-cohort nutritional effect or replace the parent’s prespecified child analysis. The original exploratory endline-due regressions and ineligible-household estimates remain in Appendix Table 17; neither establishes zero nutrition benefits or zero spillovers.

Saving and productive assets describe possible consequences beyond current food access. The Gikuriro saving coefficient in the initial exploratory specification has pointwise p -value < 0.001 and Holm p -value 0.052. It does not survive the stated economic-outcome family at five percent. A saving interpretation is therefore a hypothesis. Appendix B shows that improved returns to saving could raise intertemporal utility while reducing current food spending, and discrete food thresholds could leave variety unchanged. The experiment bundles saving groups with other activities, so its assignment cannot identify that channel or calibrate its structural parameters.

6.3 Deployment and omitted value

An allocation based on these estimates refers to villages comparable to the original trial, under its eligibility rule and within-village saturation. It assumes the same package delivery, stable proportional costs, and no response of cross-village prices or networks to the changed treatment shares. Random assignment identifies the original assignment comparisons under its design; it does not identify this scale margin. A requirement of universal assistance, geographic continuity, or a guaranteed minimum package must enter the feasible set explicitly.

If a broader objective adds unmeasured benefits H_a in the same units as measured diet value, Gikuriro exceeds a lottery q when $H_g - \sum_a q_a H_a \geq -(\mu_g - q'\mu)$. The fitted mean shortfall against lower/large cash is 0.237 diet-equivalent groups. That is a break-even accounting quantity, not

an estimate of health, saving returns, or social welfare. The full-baseline comparison can already reverse sign. Monetizing unspecified benefits cannot repair identification or select a preferred ranking.

7 Discussion and conclusion

This reanalysis leaves a difference between a plausible allocation and a defensible guarantee. Lower/large cash is the fitted mean winner at Gikuriro’s standardized expected budget. Diversifying the choice lowers worst-case measured loss within each common region. The magnitude of that protection changes sharply with inferential scope: the finite conditional bound remains wide, while the smaller block-region bound inherits documented stress-test failures. I consequently make no reliable quarter-, half-, or one-group certification claim for a deployment policy.

These results refine the economic interpretation of cash benchmarking. The line between two observed package distributions describes a village lottery; it is not a measured response to a universal intermediate cash amount. A known-mean frontier can have a two-package optimum even when a region-based choice optimally spreads aid across more packages. Failure to choose a statistically precise winner also does not imply that every diversified rule is uninformative. The confidence region, rather than a selected significance test, determines the loss calculation.

The empirical limits remain substantial. Household food access is not child nutrient intake. Endline timing differs across packages, the observation period is short, and later treatment of control limits sustained comparisons. Eligible-sample weights do not by themselves establish representativeness of unobserved households. Source assignment details leave conditional exchangeability as an explicit assumption. National-scale average costs do not identify mixed-program activation costs, and changed coverage can alter spillovers. None of these concerns is resolved by a favorable coefficient or a new transformation of the same diet data.

Further evidence could distinguish durable dietary responses, future returns, and coverage spillovers. That agenda motivates repeated diet measures, saving-group terms, and saturation variation. It does not follow as a mathematical necessity from the width of the particular outer bound reported here. The present evidence supports conditional comparisons and exposes the uncertainty an organization must face. It does not establish a general superiority of cash, a Catholic-specific mechanism, or a statistically minimax allocation rule.

References

- Banerjee, Abhijit V., and Esther Duflo. 2007. “The Economic Lives of the Poor.” *Journal of Economic Perspectives* 21(1): 141–167. doi:10.1257/jep.21.1.141.
- Currie, Janet, and Firouz Gahvari. 2008. “Transfers in Cash and In-Kind: Theory Meets the Data.” *Journal of Economic Literature* 46(2): 333–383. doi:10.1257/jel.46.2.333.
- Cunha, Jesse M., Giacomo De Giorgi, and Seema Jayachandran. 2019. “The Price Effects of Cash Versus In-Kind Transfers.” *Review of Economic Studies* 86(1): 240–281. doi:10.1093/restud/rdy018.
- Egger, Dennis, Johannes Haushofer, Edward Miguel, Paul Niehaus, and Michael Walker. 2022. “General Equilibrium Effects of Cash Transfers: Experimental Evidence from Kenya.” *Econometrica* 90(6): 2603–2643. doi:10.3982/ECTA17945.
- Food and Agriculture Organization of the United Nations. 2011. *Guidelines for Measuring Household and Individual Dietary Diversity*. Gina Kennedy, Terri Ballard, and Marie-Claude Dop. Rome: FAO. https://www.fao.org/fileadmin/user_upload/wa_workshop/docs/FAO-guidelines-dietary-diversity2011.pdf.
- Foster, James, Joel Greer, and Erik Thorbecke. 1984. “A Class of Decomposable Poverty Measures.” *Econometrica* 52(3): 761–766. doi:10.2307/1913475.
- Haushofer, Johannes, and Jeremy Shapiro. 2016. “The Short-Term Impact of Unconditional Cash Transfers to the Poor: Experimental Evidence from Kenya.” *Quarterly Journal of Economics* 131(4): 1973–2042. doi:10.1093/qje/qjw025.
- Imbens, Guido W., and Michal Kolesár. 2016. “Robust Standard Errors in Small Samples: Some Practical Advice.” *Review of Economics and Statistics* 98(4): 701–712. doi:10.1162/REST_a_00552.
- Kitagawa, Toru, and Aleksey Tetenov. 2018. “Who Should Be Treated? Empirical Welfare Maximization Methods for Treatment Choice.” *Econometrica* 86(2): 591–616. doi:10.3982/ECTA13288.
- Manski, Charles F. 2004. “Statistical Treatment Rules for Heterogeneous Populations.” *Econometrica* 72(4): 1221–1246. doi:10.1111/j.1468-0262.2004.00530.x.
- McIntosh, Craig. 2025. “Replication Package EJ MS 20220682.” Corrected dataset, created July 15, 2025, nominal publication date May 23, 2024, version 1. Zenodo. doi:10.5281/zenodo.15881329. Accessed September 30, 2026. CC BY 4.0. The record lists McIntosh as creator; the underlying paper is coauthored with Andrew Zeitlin.
- McIntosh, Craig, and Andrew Zeitlin. 2024. “Cash Versus Kind: Benchmarking a Child Nutrition Program Against Unconditional Cash Transfers in Rwanda.” *Economic Journal* 134(664): 3360–3389. doi:10.1093/ej/ueae050.
- Hoeffding, Wassily. 1963. “Probability Inequalities for Sums of Bounded Random Variables.” *Journal of the American Statistical Association* 58(301): 13–30. doi:10.1080/01621459.1963.10500830.
- Manski, Charles F. 2007. “Minimax-Regret Treatment Choice with Missing Outcome Data.” *Journal of Econometrics* 139(1): 105–115. doi:10.1016/j.jeconom.2006.06.006.
- Stoye, Jörg. 2007. “Minimax Regret Treatment Choice with Incomplete Data and Many Treatments.” *Econometric Theory* 23(1): 190–199. doi:10.1017/S0266466607070089.

A Decision and inference proofs

Proof of Proposition 1. The distribution and objective follow from conditioning on package assignment under stable delivery and no cross-village interference. The feasible set is a compact polytope. A linear objective attains its maximum at a vertex. If the cost constraint is slack, the probability simplex permits only one positive coordinate at a vertex. If it binds, three positive coordinates allow a nonzero perturbation satisfying both the sum and cost equalities. A sufficiently small positive or negative perturbation preserves nonnegativity, contradicting extremality. Thus no vertex requires more than two packages. Equal costs or an exactly affordable pure package do not alter this upper bound. Corollary 1 follows by subtracting the control mean and using $q_L = c_g/c_L$.

Proof of Proposition 2. For each μ , maximizing $(r - q)'\mu$ over r is a linear problem whose maximum occurs at a vertex. A finite maximum commutes with the supremum over $\mathcal{C}$. Each resulting supremum of affine functions is convex in q . Compactness gives continuity and existence of a minimizer. For comparator vertex r_j , linear-program duality gives

$$\sup_{A\mu \leq d} (r_j - q)'\mu = \min_{\lambda_j \geq 0: A'\lambda_j = r_j - q} d'\lambda_j.$$

The allocation program minimizes t subject to $q \in \mathcal{Q}(b)$ and these dual equalities with $d'\lambda_j \leq t$ for every comparator. Its constraints are linear jointly in q, t, λ_j . When the common true μ belongs to $\mathcal{C}$, the bound holds for every q at once, including the data-selected minimizer. This coverage implication does not optimize expected regret over sampling distributions. The code independently solves the primal adversary for every comparator to check dual values and allocation feasibility.

Finite conditional-assignment interval. Fix an arm, a primitive outcome supported on $[L, H]$, and its target $\mu = \sum_i w_i R_i(a) Y_i(a) / \sum_i w_i R_i(a)$, assuming a positive potential denominator. For complete baseline endpoints $R_i(a) = 1$. Define village residuals

$$X_j(\mu) = \sum_{i \in j} w_i R_i(a) \{Y_i(a) - \mu\}.$$

Because $\mu \in [L, H]$, every residual in block b lies in $[(L - \mu)W_b^{\max}, (H - \mu)W_b^{\max}]$, including villages with unobserved outcomes or a smaller total weight. This interval has width $(H - L)W_b^{\max}$. Its inverse-probability-scaled width is $(H - L)W_b^{\max}/p_{ba}$.

Under exchangeable quota assignment, the arm's residual sum in a block is a simple random sample of size k_{ba} from the fixed village residuals divided by p_{ba} . Theorem 4 of [Hoeffding \(1963\)](#) bounds the convex exponential moment of this without-replacement sum by the with-replacement moment. Applying the bounded-variable exponential inequality to those k_{ba} independent draws,

and multiplying across independent blocks, gives

$$\Pr \left(\left| \sum_j \frac{T_{ja} X_j(\mu)}{p_{b(j),a}} \right| > t \right) \leq 2 \exp \left\{ -\frac{2t^2}{(H-L)^2 V_a} \right\}.$$

The total residual expectation is zero since $\sum_j X_j(\mu) = 0$. Substituting (6) makes this failure probability at most α/M . The realized residual is $\hat{D}_a(\hat{\mu}_a - \mu)$. Dividing by its positive realized denominator therefore yields the reported ratio interval. This inversion tests the fixed true μ ; it needs no union over a grid of candidate means. A zero realized denominator receives full support. A union bound protects the M fixed primitive arm/transformation/endpoints. Applying deterministic endpoint ordering then covers true missing-diet means and every policy computed from the same primitive region. Dependence across arms is allowed in the union bound; independent blocks and conditional exchangeability remain essential. The bound is classical and conservative, not an optimal confidence procedure.

Block covariance and the referee counterexample. For independent population linearization scores ψ_b with means m_b satisfying $\sum_b m_b = 0$, the expectation of their centered block covariance is

$$\mathbb{E} \left[\frac{B}{B-1} \sum_b (\psi_b - \bar{\psi})(\psi_b - \bar{\psi})' \right] = \sum_b \text{Var}(\psi_b) + \frac{B}{B-1} \sum_b m_b m_b'.$$

The excess is positive semidefinite. Estimated ratio denominators and centering require an asymptotic linearization, so this identity alone proves no finite-sample coverage. In a ten-village block with five fixed scores 4 and five 6 for every arm, assign one village each to lower and upper cash without replacement. Exhausting all 90 ordered pairs gives covariance $-1/(9B)$ between the two arm means across B independent blocks. The true difference variance is $20/(9B)$; discarding cross-arm dependence instead gives $2/B$, only 90 percent of that variance. The common block vector retains the missing covariance. The synthetic coverage failures in Table 2 show why repairing this algebra is insufficient to validate the subsequent multiplier approximation with 22 blocks.

Distributional priorities. On the finite integer support, any increasing function is its value at zero plus a nonnegative combination of survival indicators. Increasing-concave functions equivalently have nonnegative weights on truncated linear gains. Thus simultaneous distributional ordering would imply the corresponding class of preferences over this proxy. The empirical regions do not establish such an ordering. Preference over HDDS is distinct from preference over nutrients or full intertemporal resources.

B An illustrative saving mechanism

B.1 Household allocation

A household has current resources $w + t$, where w is its initial budget and t is a transfer. It allocates resources to current food expenditure f and saving s . Food is measured in units with price normalized to one. Let $\underline{f}$ denote subsistence expenditure, R the positive gross return on saving, $y > 0$ background future income, and $\beta > 0$ the weight on future consumption. For an interior household with $w + t > \underline{f}$, preferences and the budget constraint are

$$U(f, s) = \log(f - \underline{f}) + \beta \log(Rs + y), \quad f + s = w + t, \quad s \geq 0. \quad (8)$$

The future resource y is important: it allows a change in the saving return to affect the allocation between dates. The model abstracts from risk, credit, household bargaining, and the multiple activities provided by Gikuriro. Its comparative statics describe a possible mechanism within that bundle.

Food diversity is an increasing step function of spending,

$$D(f) = \sum_{j=1}^{12} \mathbf{1}\{f \geq a_j\}, \quad \underline{f} \leq a_1 \leq \dots \leq a_{12}, \quad (9)$$

where a_j are household-specific thresholds for including additional food groups. This reduced form allows a household to maintain a staple diet while deferring more expensive sources of variety. It does not imply that each food group has the same nutrient content or that variety depends only on total expenditure.

Proposition 3 (Current food and saving). *If the optimum is interior, then*

$$f^* = \frac{w + t + y/R + \beta \underline{f}}{1 + \beta}, \quad s^* = \frac{\beta(w + t - \underline{f}) - y/R}{1 + \beta}.$$

Cash increases both current food expenditure and saving. A rise in R decreases current food expenditure, increases saving, and increases maximized household utility. Dietary diversity weakly increases with cash, but can remain unchanged over an interval of transfer values.

If saving is constrained at zero, a small change in its return has no effect on allocation; cash initially finances current spending. The saving-return prediction holds y fixed. I do not estimate R , y , or β : saving groups may change fees, credit, risk or commitments instead. The model illustrates how current food access and intertemporal utility can rank an intervention differently.

Corollary 2 (A short observation window). *A saving intervention can raise household utility while leaving the observed diversity score unchanged, or reducing it. A small cash transfer can raise food spending without increasing the diversity score. A larger transfer crosses a threshold only if an*

uncrossed threshold is reachable; a household already consuming all twelve groups cannot increase this score.

The bundled assignment cannot distinguish this channel from delivery timing or other program components. Repeated measurements or randomized mechanism variation would be needed.

Proof of Proposition 3. Substitution of $s = w + t - f$ gives the first-order condition $1/(f - \underline{f}) = \beta R/(Rs + y)$. Strict concavity gives the unique interior optimum stated in the proposition. Its derivatives are $\partial f^*/\partial t = 1/(1 + \beta)$, $\partial s^*/\partial t = \beta/(1 + \beta)$, and $\partial f^*/\partial R = -y/\{(1 + \beta)R^2\} < 0$. The saving derivative has the opposite sign. The envelope derivative of maximized utility is $\beta s^*/(Rs^* + y) > 0$ at an interior positive-saving optimum. At the zero-saving boundary, the allocation response to a small return change is zero. Dietary diversity is a step function; transfer increases weakly raise it and may leave it unchanged. Once all twelve thresholds are crossed, the ceiling prevents a further score gain. No R , y , or β is calibrated from the bundled reduced form.

C Regression crosswalk and supplementary estimates

For comparison with the original study, the supplementary weighted ANCOVA is

$$v(D_{i1}) = \alpha + \sum_{a \neq 0} \tau_a T_{j(i),a} + \rho v(D_{i0}) + \gamma M_i + \lambda_{b(i)} + \varepsilon_i. \quad (10)$$

The baseline transformation is replaced by its weighted within-block mean where missing, with indicator M_i ; constant regressors are omitted. Source expansion weights enter WLS. CR1 uses $G/(G-1)(N-1)/(N-K)$, with pointwise t_{G-1} intervals. The original 204-test maximum-studentized family uses 9,999 village-score Rademacher draws; these are working asymptotic bands. Secondary outcomes have 96 distinct Holm tests, food indicators 80, baseline heterogeneity ten, and observation five. Every child and ineligible sensitivity has its own fifteen-test family. These separate families do not supply one paper-wide coverage claim.

The common-slope ANCOVA need not preserve assignment-adjusted target weights under heterogeneous effects. Complete-case selection remains unresolved. The coarse all-or-nothing bounds are retained solely as the initial sample crosswalk; the main item-informed conditional regions supersede their inferential interpretation. No statistical significance is required to retain an analysis, and every referee-driven extension is exploratory.

Table 8: Initial complete-case diet ANCOVA by assignment

	Gikuriro	Lower	Middle	Upper	Large
Dietary diversity	0.255 (0.131) [0.912]	0.353 (0.182) [0.918]	−0.183 (0.228) [1.000]	0.348 (0.165) [0.804]	0.545*** (0.134) [< 0.001]
Diet-shortfall reduction ($z = 6$)	0.022 (0.016) [1.000]	0.047 (0.020) [0.581]	−0.029 (0.028) [1.000]	0.045 (0.021) [0.712]	0.073*** (0.016) [< 0.001]
Squared-shortfall reduction	0.016 (0.011) [0.999]	0.033 (0.013) [0.346]	−0.019 (0.015) [1.000]	0.026 (0.015) [0.988]	0.043*** (0.012) [0.005]
At least 4 food groups	0.018 (0.029) [1.000]	0.045 (0.037) [1.000]	−0.065 (0.044) [0.999]	0.074 (0.036) [0.837]	0.093* (0.030) [0.061]
At least 6 food groups	0.036 (0.032) [1.000]	0.055 (0.044) [1.000]	−0.045 (0.078) [1.000]	0.068 (0.058) [1.000]	0.119* (0.038) [0.053]
At least 8 food groups	0.038 (0.024) [0.996]	0.047 (0.036) [1.000]	−0.017 (0.026) [1.000]	0.014 (0.029) [1.000]	0.029 (0.026) [1.000]
Households (range across outcomes)	1728	1728	1728	1728	1728
Villages (range across outcomes)	248	248	248	248	248

Notes: Assignment effects, baseline transformation, missing-baseline indicator, blocks, and source weights. Parenthesized SEs are village CR1. Displayed simultaneous corrections are the original working asymptotic family, not validated full-baseline bounds.

Table 9: Initial ANCOVA comparisons at the Gikuriro budget

	Control + large	Lower + Large	Middle + Large	Upper + Large
Dietary diversity	0.124 (0.119) [1.000]	-0.127 (0.162) [1.000]	0.389 (0.213) [0.963]	-0.094 (0.170) [1.000]
Shortfall reduction ($z = 4$)	0.006 (0.011) [1.000]	-0.017 (0.013) [1.000]	0.035 (0.016) [0.746]	-0.009 (0.017) [1.000]
Shortfall reduction ($z = 6$)	0.004 (0.014) [1.000]	-0.029 (0.018) [0.997]	0.044 (0.026) [0.990]	-0.024 (0.021) [1.000]
Shortfall reduction ($z = 8$)	0.014 (0.014) [1.000]	-0.018 (0.019) [1.000]	0.046 (0.026) [0.983]	-0.013 (0.021) [1.000]
Squared-shortfall reduction	0.006 (0.010) [1.000]	-0.019 (0.012) [0.996]	0.031 (0.015) [0.818]	-0.010 (0.015) [1.000]
At least 6 food groups	0.008 (0.030) [1.000]	-0.029 (0.040) [1.000]	0.070 (0.072) [1.000]	-0.032 (0.058) [1.000]
Households (range across outcomes)	1728	1728	1728	1728
Villages (range across outcomes)	248	248	248	248

Notes: Positive values favor Gikuriro over the named cash lottery. Coefficients and village CR1 SEs use the initial complete-case ANCOVA. Inference and target composition differ from the main conditional baseline calculation.

Table 10: Source economic outcomes: exploratory assignment regressions

	Gikuriro	Lower	Middle	Upper	Large
Consumption (IHS)	−0.093 (0.099) [1.000]	0.118 (0.116) [1.000]	−0.001 (0.128) [1.000]	0.083 (0.120) [1.000]	0.327 (0.108) [0.245]
Productive assets (IHS)	0.008 (0.103) [1.000]	0.271 (0.127) [1.000]	0.248 (0.175) [1.000]	0.246 (0.135) [1.000]	0.780*** (0.118) [< 0.001]
Saving stock (IHS)	1.186* (0.338) [0.052]	−0.009 (0.447) [1.000]	−0.286 (0.539) [1.000]	0.353 (0.525) [1.000]	0.720 (0.322) [1.000]
Borrowing stock (IHS)	0.056 (0.358) [1.000]	−0.667 (0.339) [1.000]	−0.714 (0.434) [1.000]	−0.830 (0.375) [1.000]	−0.325 (0.388) [1.000]
Health knowledge index	−0.062 (0.366) [1.000]	0.273 (0.488) [1.000]	0.041 (0.410) [1.000]	0.068 (0.373) [1.000]	0.039 (0.474) [1.000]
Sanitation practices index	−0.207 (0.227) [1.000]	0.337 (0.363) [1.000]	−0.229 (0.286) [1.000]	0.323 (0.283) [1.000]	0.112 (0.298) [1.000]
Purchased food (IHS)	−0.172 (0.177) [1.000]	0.058 (0.192) [1.000]	0.024 (0.187) [1.000]	0.264 (0.188) [1.000]	0.267 (0.170) [1.000]
Own-produced food (IHS)	0.125 (0.322) [1.000]	0.556 (0.378) [1.000]	−0.601 (0.427) [1.000]	−0.558 (0.556) [1.000]	0.968 (0.393) [1.000]
Households (range across outcomes)	1751	1751	1751	1751	1751
Villages (range across outcomes)	248	248	248	248	248

Notes: IHS coefficients are transformation units, not percentage changes. Source preprocessing is inherited except the stated food-value IHS transformation. Village CR1 SEs and the separate 96-test economic Holm family. No causal mediation or total-welfare interpretation.

Table 11: Follow-up child scores in the fixed baseline-measured cohort

Outcome	Gikuriro	Lower	Middle	Upper	Large
Height-for-age	0.042 (0.050)	0.031 (0.062)	-0.080 (0.077)	-0.038 (0.061)	0.090 (0.056)
Holm p	1.000	1.000	1.000	1.000	1.000
N	2141	2141	2141	2141	2141
Weight-for-age	0.042 (0.040)	0.008 (0.056)	-0.019 (0.057)	0.041 (0.043)	0.081 (0.039)
Holm p	1.000	1.000	1.000	1.000	0.566
N	2131	2131	2131	2131	2131
Arm circumference	0.024 (0.056)	-0.029 (0.078)	0.009 (0.107)	-0.070 (0.067)	0.140 (0.078)
Holm p	1.000	1.000	1.000	1.000	1.000
N	1837	1837	1837	1837	1837

Notes: Membership fixed by baseline eligibility and source anthropometry flag, regardless of endline eligibility/due status. Regression uses available follow-up scores, original assignment/weights, baseline source score, missing-lag indicator, and blocks. No full-cohort selection-robust effect is claimed. Village CR1 SEs; fifteen-test Holm family separate from the endline-due analysis.

Table 12: Food-module information in the baseline eligible cohort

Arm	Baseline	Identified	Partial	Unobserved
Control	521	498	6	17
Gikuriro	540	522	2	16
Lower	165	160	0	5
Middle	154	149	0	5
Upper	167	164	3	0
Large	246	237	4	5

Notes: Identified includes complete modules and partial modules that nevertheless identify all twelve groups. Partial denotes an interval of width between one and eleven. Unobserved denotes width twelve, including absent endline households. Every baseline eligible household occurs once. These counts are not a missing-at-random adjustment.

Table 13: Crosswalk from the original diet specification

Specification	GK	GK SE	Large	Large SE	N
Original controls, pooled cash	0.192	0.122	0.550	0.126	1751
Original controls, split cash	0.201	0.123	0.558	0.126	1751
Supplied-score lag only	0.235	0.131	0.530	0.131	1751
Integer outcome, supplied lag	0.258	0.130	0.547	0.133	1728
Integer outcome and baseline	0.255	0.131	0.545	0.134	1728

Notes: Original controls reproduce the fixed covariate list in the corrected deposit’s `CovariateLists.xlsx` and `Table_3.do`, including the supplied lag. Pooled means the three smaller cash arms share a coefficient. Split estimates them separately. Collinear supplied baseline columns are omitted in the stated order, matching the identified regression span. Later rows change the control list and integer-score missingness sequentially. Standard errors use village CR1. The final row is this project’s original ANCOVA. Published rounding, rather than an exact recovery of a withdrawn dataset, is the comparison.

Table 14: Mean-diet small-cluster sensitivity

Contrast	Effect	CR2 SE	Satt. df	Pointwise p	Effective clusters
Gikuriro	0.255	0.139	91.8	0.069	28.0
Lower	0.353	0.196	28.8	0.083	10.1
Middle	-0.183	0.256	21.2	0.482	3.9
Upper	0.348	0.180	26.2	0.065	10.5
Large	0.545	0.142	47.9	< 0.001	34.4
GK vs. Control + Large	0.124	0.126	96.7	0.327	27.8
GK vs. Lower + Large	-0.127	0.175	34.6	0.474	12.9

Notes: Same WLS coefficient specification as the initial ANCOVA; CR2 changes its variance estimate. Satterthwaite degrees of freedom use the independent, homoskedastic whitened-error working model. Effective clusters are the inverse concentration of squared cluster scores for the indicated contrast. These are pointwise diagnostic p -values, not the primary simultaneous policy family or exact randomization tests. Working-model sandwich expectation identities are verified in code.

Table 15: Village weight support among complete diets

Arm	Villages	Effective weight villages	Maximum village weight (%)
Control	74	44.6	5.4
Gikuriro	74	51.7	5.6
Lower	22	16.7	10.7
Middle	22	12.6	17.9
Upper	22	15.0	15.1
Large	34	24.1	8.9

Notes: Village weights sum source sampling weights over observed eligible diets. Effective support is $(\sum_j W_j)^2 / \sum_j W_j^2$. It concerns weight concentration, not the contrast-specific score concentration in Table 14.

Table 16: Omitting each village and each randomization block

Omitted unit	Cash comparator	Minimum effect	Maximum effect	Fits
Village	Control + Large	0.087	0.172	248
Village	Lower + Large	-0.209	-0.058	248
Village	Upper + Large	-0.143	0.019	248
Block	Control + Large	0.068	0.171	22
Block	Lower + Large	-0.219	-0.038	22
Block	Upper + Large	-0.137	-0.006	22

Notes: Minimum and maximum reestimated Gikuriro-minus-policy mean coefficients. Each fit recomputes baseline imputation, block indicators and covariance on the remaining data, holding original package costs fixed. Omission ranges are sensitivity diagnostics, not confidence intervals.

Table 17: Available child growth and ineligible-household outcomes

Outcome	GK	SE	Holm p	Large	SE	Holm p	N
Child height-for-age	0.004	0.057	1.000	0.086	0.069	1.000	2607
Child weight-for-age	-0.003	0.040	1.000	0.092	0.045	0.608	2593
Child arm circumference	-0.023	0.053	1.000	0.148	0.073	0.616	2162
Ineligible HDDS	0.111	0.142	1.000	-0.392	0.209	0.798	960
Ineligible consumption (IHS)	-0.149	0.118	1.000	-0.154	0.201	1.000	967
Ineligible saving (IHS)	-0.334	0.423	1.000	-0.931	0.455	0.629	967

Notes: Source-weighted assignment effects with block/baseline controls and village-clustered SE. Child standardized scores use the source endline due-for-measurement sample; ineligible households use baseline eligibility. Separate fifteen-test Holm families include all five arms; counts differ by outcome. These selected exploratory samples do not replace the parent paper’s richer prespecified anthropometric regressions.

Table 18: Baseline characteristics of eligible households

	Control	Gikuriro	Lower	Middle	Upper	Large
Baseline diet (groups)	4.25 (2.01)	4.14 (1.87)	4.29 (2.16)	3.92 (1.85)	4.23 (1.92)	4.13 (1.88)
Consumption (IHS)	10.45 (1.42)	10.49 (1.49)	10.38 (1.64)	10.53 (1.29)	10.49 (1.37)	10.33 (1.43)
Household members	5.27 (1.85)	5.18 (1.80)	5.29 (1.95)	5.37 (2.06)	4.98 (1.75)	5.08 (1.64)
Female head (share)	0.13 (0.34)	0.17 (0.38)	0.17 (0.37)	0.15 (0.35)	0.21 (0.41)	0.12 (0.33)

Notes: Baseline survey-weighted means with weighted population standard deviations in parentheses. Diet uses the reconstructed complete food module. Each variable is summarized over its nonmissing baseline observations. The replication output reports those counts and a separate set of block-adjusted balance tests.

Table 19: Feasible cash/control vertices at the Gikuriro budget

Cash policy	Expected USD	Cash coverage (%)	Large-cash share (%)
Control	0.00	0.00	0.00
Lower	53.58	100.00	0.00
Middle	95.86	100.00	0.00
Upper	121.24	100.00	0.00
Control + Large	124.49	24.06	24.06
Lower + Large	124.49	100.00	15.29
Middle + Large	124.49	100.00	6.79
Upper + Large	124.49	100.00	0.82

Notes: Coverage denotes assignment to any cash package, not verified household take-up. Lotteries operate on whole villages. Expected cost is per eligible household at original accounting costs. Pure lower, middle, and upper policies leave resources unspent.

Table 20: Sensitivity of mean-diet contrasts

	Control + large	Upper + large
Block controls only	0.111 (0.130)	-0.110 (0.170)
Equal household weights	0.153 (0.109)	0.051 (0.178)
Equal village weights	0.134 (0.116)	0.037 (0.180)
Supplied diet score	0.107 (0.118)	-0.104 (0.162)
Observed baseline diet	0.139 (0.119)	-0.091 (0.173)

Notes: Each entry is Gikuriro minus the indicated cash policy. Standard errors clustered by village are in parentheses. All specifications retain block fixed effects. Equal-village weighting normalizes survey weights to sum to one within each observed village. Complete-baseline analysis excludes missing baseline food modules; supplied-score analysis keeps the original replacements. These sensitivity estimates are not selected for significance.

Table 21: Sample identification regions allowing arbitrary missing diets

Outcome	Cash comparison	Lower bound	Upper bound
Diet	Control	-0.263	0.533
Diet	Lower	-0.555	0.188
Diet	Middle	0.215	0.852
Diet	Upper	-0.330	0.111
Diet	Control + Large	-0.351	0.417
Diet	Lower + Large	-0.566	0.167
Diet	Middle + Large	0.158	0.798
Diet	Upper + Large	-0.333	0.111
Shortfall reduction	Control	-0.018	0.049
Shortfall reduction	Lower	-0.059	0.003
Shortfall reduction	Middle	0.038	0.091
Shortfall reduction	Upper	-0.049	-0.012
Shortfall reduction	Control + Large	-0.032	0.032
Shortfall reduction	Lower + Large	-0.061	-0.000
Shortfall reduction	Middle + Large	0.030	0.084
Shortfall reduction	Upper + Large	-0.049	-0.012

Notes: Worst-case bounds use the full baseline eligible sample and baseline sampling weights. Score bounds are 0 and 12; negative normalized-shortfall bounds are -1 and 0 . The bounds have no adjustment for block composition or sampling uncertainty. They are sample identification regions, not confidence intervals or evidence of a significant policy ranking.

Table 22: Baseline heterogeneity in package diet impacts

Arm	Low baseline diet	Low baseline consumption
Gikuriro	-0.531 (0.281)	-0.149 (0.248)
Holm p	0.600	1.000
Lower	0.131 (0.338)	-0.262 (0.288)
Holm p	1.000	1.000
Middle	-0.320 (0.382)	-0.588 (0.582)
Holm p	1.000	1.000
Upper	-0.281 (0.437)	-0.204 (0.360)
Holm p	1.000	1.000
Large	-0.408 (0.331)	-0.104 (0.258)
Holm p	1.000	1.000

Notes: Coefficients are low-minus-high differences in HDDS ITT. Low status is defined strictly below the median of observed baseline diet or source consumption-IHS. Interactions are included for every package; sample selection uses only the observed baseline split variable. Standard errors are in parentheses. Holm adjustment covers all ten interaction tests. Thresholds and observations are stored in the machine-readable output.

Table 23: Large-cash effects on individual food categories

Food category	Large-cash ITT	SE	Holm p
Cereals	0.123	0.028	0.001
Roots and tubers	-0.022	0.051	1.000
Vitamin A vegetables	-0.045	0.028	1.000
Leafy vegetables	0.005	0.041	1.000
Other vegetables	0.026	0.045	1.000
Vitamin A fruits	0.061	0.030	1.000
Other fruits	0.010	0.028	1.000
Organ meat	0.010	0.013	1.000
Other meat	-0.019	0.012	1.000
Eggs	-0.006	0.016	1.000
Fish	0.042	0.030	1.000
Legumes	0.064	0.028	1.000
Milk	-0.005	0.026	1.000
Oils	0.158	0.035	< 0.001
Sweets	0.052	0.034	1.000
Spices and condiments	0.055	0.038	1.000

Notes: Outcomes are binary category responses in the original 16-category module, so vegetable, fruit, and meat subcategories remain separate here. Responses outside $\{0, 1\}$ are missing. Effects are changes in probability, with weighted ANCOVA and village-clustered standard errors. Holm adjustment covers 16 categories and five treatment arms, including estimates not displayed in this table. These are household consumption reports, not measured child nutrients.

Table 24: Assignment effects on observing an endline diet

Arm	Observed-diet effect	SE	Holm p
Gikuriro	1.29	1.04	0.875
Lower	0.32	1.60	1.000
Middle	1.48	1.47	0.953
Upper	3.11	1.10	0.026
Large	0.72	1.27	1.000

Notes: Effects and standard errors are in percentage points. The outcome is one if a baseline eligible household is in the endline panel with a complete food module, and zero otherwise. Estimates use baseline sampling weights, block fixed effects and village-clustered standard errors. Holm adjustment covers the five arm-versus-control observation tests. This is a distinct diagnostic family; no observation effect proves ignorability of missing diet.

D Replication and verification

The public package contains a single master script, frozen independent input hashes, corrected source attribution, codebooks, extraction programs, dependency locks, numerical outputs, generated LaTeX tables and in-text macros, and the compiled PDF. The source ZIP is excluded; its numeric research extract is redistributed under CC BY 4.0. Recreating this extract from the verified archive is distinct from recreating the parent’s entire upstream survey processing.

The initial mean ANCOVA is checked by an independent Stata implementation. Scientific invariants, policy-vertex comparisons against independent linear programs, bounded-sampling exponential moments, block-score algebra, sparse-outcome support, cohort partitions, and primal/dual adversary values are checked separately. The master retains failed synthetic coverage results; a successful implementation check cannot turn them into passed coverage tests. Cold-output and separate-environment comparisons record their actual source version and scope in the public audit. They do not imply independent data-editor replication.

The JPE-template report is a self-audit, rather than an official journal report. Genuine ChatGPT referee text and point-by-point responses are preserved separately. Simulated referee recommendations are not real journal acceptance. The original secondary plan and all amendments remain available, including the replacement of the unsupported village-score certificates by the stated finite conditional outer regions and exploratory block sensitivity.